Example 2.65

Evaluate $\int_{-1}^{1} x\sqrt{x+1}\ dx$

Solution:

Take $t = x+1$ $dt = dx$ and		
x	-1	1
t	0	2

$$\int_{-1}^{1} x\sqrt{x+1}\ dx = \int_{0}^{2} (t-1)\sqrt{t}\ dt$$

$$= \int_{0}^{2} \left(t^{\frac{3}{2}} - t^{\frac{1}{2}} \right) dt$$

$$= \left[\frac{2t^{\frac{5}{2}}}{5} - \frac{2t^{\frac{3}{2}}}{3} \right]_0^2$$

$$= \frac{8\sqrt{2}}{5} - \frac{4\sqrt{2}}{3}$$

$$= \frac{4\sqrt{2}}{15}$$

Example 2.66

Evaluate $\int_{0}^{\infty} e^{-\frac{x}{2}}\ dx$

Solution:

$$\int_{0}^{\infty} e^{-\frac{x}{2}}\ dx = -2\left[e^{-\frac{x}{2}} \right]_0^{\infty}$$

$$= -2[0-1] = 2$$

Example 2.67

Evaluate $\int_{0}^{\infty} x^2 e^{-x^3}\ dx$

Solution:

Take $x^3 = t$ $3x^2\ dx = dt$ $\Rightarrow x^2\ dx = \frac{dt}{3}$ and		
x	0	∞
t	0	∞

$$\int_{0}^{\infty} x^2 e^{-x^3}\ dx = \int_{0}^{\infty} e^{-t}\ \frac{dt}{3}$$

$$= \frac{1}{3}\left[-e^{-t} \right]_0^{\infty}$$

$$= \frac{-1}{3}[0-1]$$

$$= \frac{1}{3}$$

Example 2.68

Evaluate $\int_{1}^{2} \frac{1}{(x+1)(x+2)} dx$

Solution:

By partial fractions,

$$\frac{1}{(x+1)(x+2)} = \frac{A}{x+1} + \frac{B}{x+2}$$

$$\Rightarrow \frac{1}{(x+1)(x+2)} = \frac{1}{x+1} - \frac{1}{x+2}$$

$$\int_{1}^{2} \frac{1}{(x+1)(x+2)} dx = \int_{1}^{2} \left[\frac{1}{x+1} - \frac{1}{x+2} \right] dx$$

$$= \left[\log|x+1| - \log|x+2| \right]_1^2$$

$$= \log\frac{3}{4} - \log\frac{2}{3}$$

$$= \log\frac{9}{8}$$

Example 2.69

Evaluate $\int_{1}^{e} \log x\ dx$

Solution:

Take $u = \log x$ Differentiate $du = \frac{1}{x} dx$	$dv = dx$ Integrate $v = x$

$$\int_{1}^{e} \log x\ dx = \int_{1}^{e} u\ dv$$

$$= [uv]_1^e - \int_{1}^{e} v\,du$$

$$= [x\log x]_1^e - \int_{1}^{e} x\frac{1}{x}\ dx$$

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